redrob | H2S HACK2SKILL

INDIA.RUNS

Build what next India runs on

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Problem Statement : Intelligent Candidate Ranking System (Founding AI Engineer)

- What is your proposed solution?
- What differentiates your approach from traditional candidate matching systems?

- What are the key requirements extracted from the JD?
- Which candidate signals are most important for determining relevance? / How does your solution evaluate candidate fit beyond keyword matching?

- How does your system retrieve, score, and rank candidates?
- What models, algorithms, or heuristics are used?
- How are multiple candidate signals combined into a final ranking?

- How are ranking decisions explained?
- How do you prevent hallucinations or unsupported justifications?
- How does your solution handle inconsistent, low-quality, or suspicious profiles?

Pipeline Workflow Sequence:

1. JD Ingestion: Recruiter enters JD text and adjusts scoring weights.

- What is the complete workflow from JD input to ranked candidate output?

2. Phase 1 Extraction: Processes raw JSONL profiles, extracts signals, and stores in features.parquet.

3. Honeypot Shield Filter: Evaluates Heuristic checks and caps suspicious profile scores at 0.05.

4. Scoring & Rank Tie-Breaking (Phase 2): Evaluates candidates, applies micro-tiebreakers, and ranks.

5. LLM Reasonings: Groq generates bespoke, compliant recommendation summaries for the top 100.

6. Sandbox Dashboard: Displays visual profiles, allows detail audits, and exports final CSV.

CANDIDATE RANKING
MACHINE LEARNING PIPELINE ARCHITECTURE

PHASE 1: FEATURE EXTRACTION
(OFFLINE/BATCH)

JOB DESCRIPTION (JD)

100K CANDIDATES.JSONL

RAW DATA

FEATURE EXTRACTION ENGINE

SEMANTIC / TF-IDF EMBEDDINGS

CAREER QUALITY

NOTICE PERIOD

LOCATION SIGNALS

PROCESSED CANDIDATE FEATURES
(PARQUET FILE)

SAVES TO

PERIODIC UPDATE

PHASE 2: REAL-TIME RANKING
(ONLINE/ON-DEMAND)

PROCESSED CANDIDATE FEATURES
(PARQUET FILE)

API REQUEST
(Candidate ID)

LOADS FROM

RANKING ENGINE

WEIGHTS CONFIGURATION

5-LAYER HONEYPOT SHIELD FILTER
PREVENTING MALICIOUS BOTTING/SCAM PROFILES

MICRO-TIEBREAKERS

OUTPUT

RANKED CANDIDATES LIST

<0.5S LATENCY

Ranking Quality Results:

- Zero honeypots or overqualified candidates in the top 10.
- What results or insights demonstrate ranking quality?
- Top ranks represent premium product engineers from Swiggy, Paytm, Amazon, Microsoft, and Razorpay.
- How does your solution meet the challenge's runtime and compute constraints?
- Outsourcing/BPO backgrounds (e.g. Genpact, Accenture, TCS) are successfully filtered out.

Computing & Constraint Performance:

- Offline Feature Extraction (Phase 1) completed in ~4 minutes.
- Real-time Inference & Ranking (Phase 2) executes in under 0.2 seconds on CPU.

- What technologies, frameworks, and tools were used and why were they selected for this solution?

Core Technology Stack:

- Pandas & PyArrow: For highly optimized Parquet read/write operations and schema enforcement.
- Scikit-Learn: For fast TF-IDF text vectorization and cosine similarity calculations.
- Streamlit: For building the premium dark-mode recruiter sandbox dashboard.
- Groq API (Llama 3.1 8B): For extremely rapid and instruction-compliant reasoning generation.
- win32com & python-pptx: For programmatic slide generation and headlessly rendering the final PDF.

Primary Submission Assets:

- GitHub Repository: https://github.com/Mohammadsiraj07/Redrob_Hackathon

Github video etc

- HuggingFace Space: https://huggingface.co/spaces/sirajmohammad6359/Redrob-Hackathon

- Key Deliverables: Ranked candidate list (submission.csv), feature store (features.parquet), presentation deck (presentation_deck.pptx & presentation_deck.pdf), and metadata (submission_metadata.yaml)

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THANK YOU